Saptaparno Chakraborty

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Skills

- **Languages:** C++, C, JavaScript, Java, Python, HTML, CSS
- **Frameworks/Libraries:** Node.js, Express, Tailwind CSS, React
- **Databases:** MySQL, SQLite, MongoDB, PostgreSQL
- **Tools/Platforms:** Git, GitHub, Linux, Postman
- **Soft Skills:** Curiosity, Problem-Solving, Discipline, Grit

Projects

WorkshopHub | Link — Nov' 25 - Present

- Designed and implemented 20+ RESTful API endpoints using Node.js and Express.js across authentication, workshops, registrations, attendance, feedback, materials, and certificate management.
- Implemented JWT-based authentication and role-based access control for 3 user roles (Admin, Instructor, Participant), using bcrypt for secure password hashing and securing 100% of protected routes.
- Modeled 7 scalable MongoDB data schemas using Mongoose, defining relationships to support the complete workshop lifecycle.
- Built end-to-end backend workflows spanning workshop creation, registrations, attendance tracking, material distribution, feedback collection, and certificate issuance.
- Developed a responsive management interface using React.js and Tailwind CSS (Vite) for role-aware interactions.
- **Tech:** Node.js, Express.js, MongoDB, Mongoose, React.js, Tailwind CSS, JWT, bcrypt

Priorify | Link — Jul' 25 - Present

- Built a deterministic, priority-driven CLI task manager in C++, supporting 6 core operations (add, update, delete, list, complete, clear) for fast local execution.
- Implemented priority ordering using std::priority_queue with a custom min-heap comparator, ensuring tasks are always displayed in ascending priority order.
- Architected a heap → vector → heap rebuild workflow using std::vector, enabling indexed task edits (update/delete/complete) and keeping ordering consistent after mutations.
- Added SQLite-based data persistence with prepared statements, ensuring 100% data retention across restarts and full task lifecycle synchronization.
- Designed as a CLI-first alternative to GUI-heavy task managers, prioritizing speed and determinism at the core, with a planned native Qt-based GUI built on the same C++ base.
- **Tech:** C++, Standard Template Library (C++), SQLite, Qt (GUI upcoming)

Training

CodeQuest: DSA Summer Bootcamp - From Basics to Brilliance — Jun' 25 - Jul' 25

Data Structures and Algorithms

- Focused on core and advanced Data Structures & Algorithms.
- Covered arrays, linked lists, stacks, queues, trees, heaps, graphs, recursion, and dynamic programming.
- Emphasized algorithmic optimization, problem-solving, and real-world software applications.
- Included daily hands-on coding, debugging, and complexity analysis sessions.

Certificates

- Computational Theory: Language Principle & Finite Automata Theory (Infosys Springboard) | Certificate — Aug' 25
- Privacy and Security in Online Social Media (NPTEL) | Certificate — Apr' 25
- Fundamentals of Network Communication (University of Colorado, Coursera) | Certificate — Nov' 24
- Introduction to Hardware and Operating Systems (IBM, Coursera) | Certificate — Sep' 24
- The Bits and Bytes of Computer Networking (Google, Coursera) | Certificate — Sep' 24
- Responsive Web Design (freeCodeCamp) | Certificate — Nov' 23

Achievements

- Leetcode - Solved 150+ Problems | Profile Link — Feb' 26
- HackerRank C++ - Gold Badge (5 Stars) | Profile Link — Nov' 25

Education

Lovely Professional University — Punjab, India
Bachelor of Technology (Computer Science and Engineering) **CGPA: 7.87** — *Aug' 23 - Present*

Railway H.S. School — West Bengal, India
Higher Secondary (XII) **Percentage: 78** — *Apr' 22 - Mar' 23*

Railway H.S. School — West Bengal, India
Secondary (X) **Percentage: 89** — *Apr' 20 - Mar' 21*